NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Interpreting a Chi-Square Result

AP Statistics · Unit 3 · Topic 3.15 · B: 2027-01-14 · E: 2027-01-15

[STAR] TODAY'S PROBLEM

Suppose a city library takes an SRS of 240 of its 6,000 active adult cardholders. It records each person's chosen notification method and whether the person returned an item late in the previous six months.

Nia: "Email/Late has the largest raw gap, 13.75; I think that cell drives the statistic and the result shows email prevents late returns."

Omar: "Expected counts differ, so raw gaps may not settle cell influence. I would calculate contributions and examine how notification methods were obtained before interpreting."

Late returns

Method	Late	Not late	Total
Email	15	85	100
Text	24	56	80
None	30	30	60
Total	69	171	240

FIRST TAKE — before we discuss (5 min)

Can this study show that choosing email causes fewer late returns? Give one reason from how the study was done.

[BOOK] SCALE EACH GAP, THEN BOUND THE CONCLUSION (keep on the page)

For independence, H_0 says the two categorical variables are not associated in the population; H_a says they are associated. Use $E =$ (row total)(column total)/(table total), and add cell contributions $(O - E)^2/E$ to obtain χ^2. With r rows and c columns, excluding totals, $df = (r-1)(c-1)$. The p-value is the upper-tail probability assuming H_0. Reject H_0 if $p \leq \alpha$; otherwise fail to reject. Random sampling supports population claims; random assignment supports causation.

Work (a)–(c) from the rules box:

DOK 1 **(a)** State the degrees of freedom, excluding the totals row and column.

DOK 2 **(b)** Expected counts are 28.75 for Email/Late and 17.25 for None/Late. Calculate these two contributions using $(O - E)^2/E$.

DOK 3 **(c)** Technology gives $\chi^2 = 22.5172$ and $p = 0.0000129$. In 3–4 sentences, judge the accounts: compare the two contributions, use $\alpha = 0.05$ for a population conclusion, and explain why the study does not show prevention.

Frame: The two contributions are __________ and __________, so ____________________ adds more to the statistic.

Frame: The p-value supports ____________________ for ____________________. This does not establish cause because ________________________.

EXIT — turn this in

One thing the rules box changed about my first take: ______________________________